

From Swept Contact to Pose: Probe-Aware Registration via Complementary-Shape Docking

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Abstract—Contact-based registration often converts each probe pose into a measured surface point and then aligns the resulting point sets. This abstraction discards the probe’s finite shape and orientation, while turning samples acquired when the probe is not in contact into false surface correspondences. We instead retain probe geometry explicitly in the measurement model: the full pose sequence and the rigid probe define a swept volume, turning registration into complementary-shape docking on $SE(3)$. Boundary agreement represents contact, while the swept volume imposes non-penetration constraints. Non-contact poses therefore remain informative without producing point residuals, naturally improving robustness to intermittent contact. Our coarse-to-fine solver combines 3D cross-correlation over translation with low-discrepancy sampling on $SO(3)$, followed by continuous pose refinement on $SE(3)$ using signed-distance sensitivities. No point correspondences or contact-state labels are required. Across five diverse object meshes, the mean translation and rotation errors were 0.031 mm and 0.325° . The method remained accurate when 98 % of the recorded poses were non-contact, and achieved errors of 0.42 mm and 3.75° in robotic tooth preparation.

I. INTRODUCTION

Registration of a prior object model to the robot frame is a prerequisite for precise manipulation. Active contact provides a direct route to this registration by sweeping a rigid probe over the object and recording its pose trajectory through robot proprioception.

Existing contact registration has largely inherited a point-measurement abstraction from coordinate metrology. Classical systems offset a measured point along a model normal to compensate for a spherical stylus [1], [2]. More recent methods align sparse touches through correspondences [3], reason probabilistically over object pose [4], [5], or enumerate hypotheses that satisfy contact and free-space constraints [6], [7]. Whenever probe poses are reduced to points, this abstraction discards orientation-dependent constraints from cylindrical or otherwise anisotropic tools and introduces a data-association problem: a sample recorded while the probe is not in contact has no corresponding object point and becomes an outlier if treated as contact.

Our key idea is simple: *a contact trajectory is a geometric object*. The union of the posed probe bodies forms a swept volume whose interior must remain disjoint from the object, while actual contacts place part of the reachable object surface on its boundary. Non-contact poses remain informative

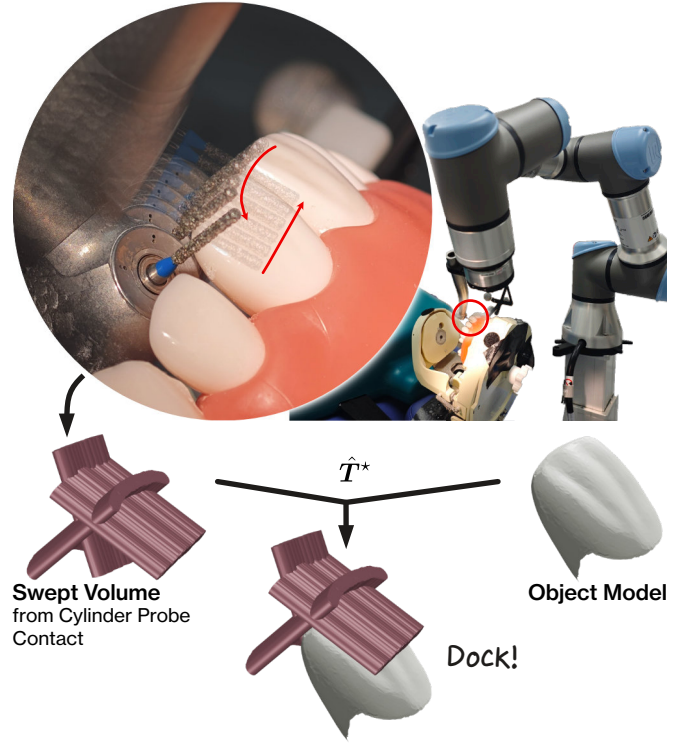


Fig. 1. Probe motion as a geometric measurement. The known probe and its pose trajectory form a swept volume whose complementary docking with the object model estimates the object pose. The tooth-preparation experiment illustrates the method on a robot.

because they impose non-penetration constraints, which naturally improves robustness to intermittent contact.

The unknown object therefore docks against the swept volume through shape complementarity, analogous to geometric docking in molecular recognition [8]. Related Fourier fields have guided arbitrary-shape assembly [9], whereas in our setting robot motion generates one of the two complementary shapes. The resulting representation preserves probe geometry, combines contact with free space, and exposes the $SE(3)$ structure of the estimation problem.

We therefore treat probe geometry and motion as part of the measurement model. We contribute (i) a representation that lifts a pose trajectory into the swept geometry of a finite rigid body, (ii) a global search over $SE(3)$ using complementary-shape matching, grid correlation, and structured rotation sampling, and (iii) continuous pose refinement through signed-distance sensitivities and $se(3)$ updates. Simulation and robotic tooth preparation evaluate this representation under geometric variation and intermittent contact.

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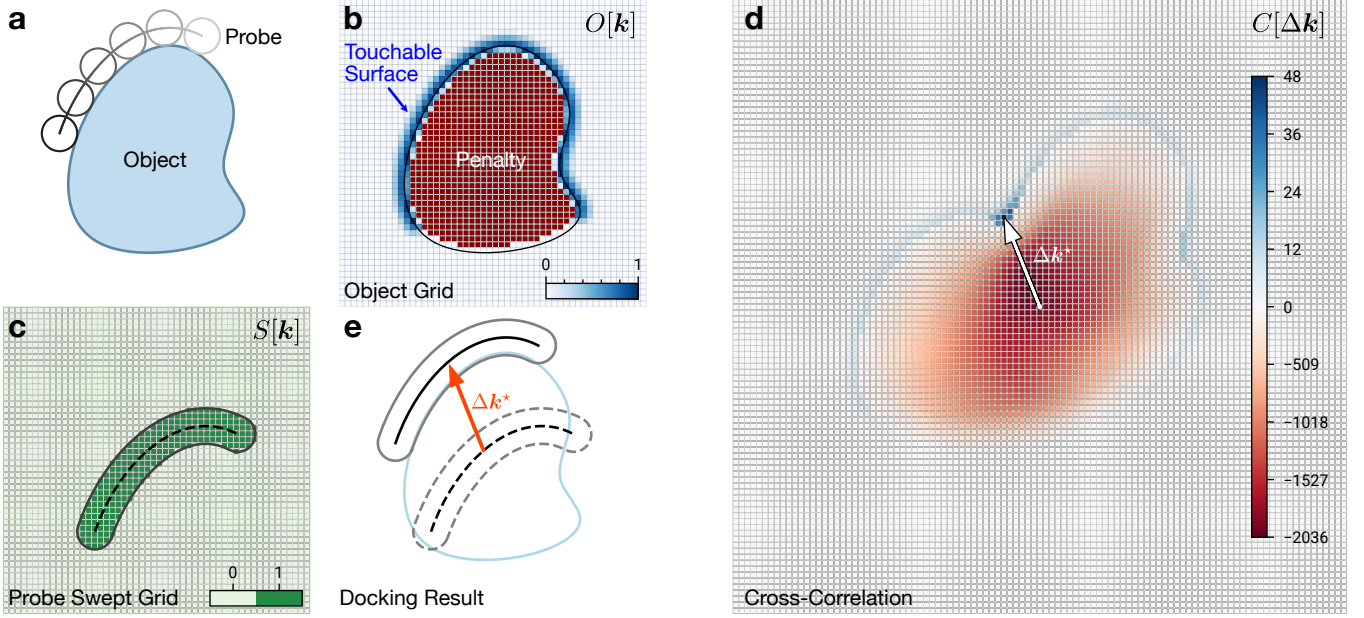


Fig. 2. Contact-to-geometry construction (shown in 2D). (a) A finite probe follows a contact trajectory. (b) The object field rewards its touchable boundary and penalizes its interior. (c) Union of all posed probes forms the swept occupancy. (d) Correlation evaluates every translation at once. (e) The maximizer docks the object against the complementary swept shape.

II. COMPLEMENTARY GEOMETRY AND GRID CORRELATION

A. The swept probe is the measurement

Let $\mathcal{B} \subset \mathbb{R}^3$ be the solid object model, $\Gamma \subset \partial\mathcal{B}$ its reachable (touchable) surface, and $\mathcal{Q} \subset \mathbb{R}^3$ the rigid probe body in its local frame. The robot records poses $\mathcal{T} = \{\mathbf{T}_i\}_{i=1}^N$, $\mathbf{T}_i \in SE(3)$. We form the swept set from the full posed probe bodies [10]

$$\mathcal{S}(\mathcal{T}, \mathcal{Q}) = \bigcup_{i=1}^N \mathbf{T}_i \mathcal{Q}. \quad (1)$$

For an unknown object pose $\mathbf{X} \in SE(3)$, ideal rigid contact imposes complementary constraints for a nonempty contacted subset $\Gamma_c \subseteq \Gamma$:

$$\begin{aligned} \text{int}(\mathbf{X}\mathcal{B}) \cap \text{int}(\mathcal{S}) &= \emptyset, \\ \mathbf{X}\Gamma_c &\subseteq \partial\mathcal{S}. \end{aligned} \quad (2)$$

The non-penetration constraint applies to the entire swept set, whereas the boundary constraint applies only to the contacted surface. Thus every pose imposes non-penetration, while contact poses additionally constrain the contacted object surface to $\partial\mathcal{S}$. No surface correspondence is assigned to a non-contact sample, so contact loss does not create a point residual.

B. Fields for contact and non-penetration

Let $\phi_{\mathcal{B}}$ be the object's signed distance field, with negative values inside the object. We map it to a narrow-band function ψ that peaks at the zero level set, tapers to zero outside an approach band, tolerates penetration at the discretization scale, and becomes strongly negative deeper inside. A narrow-band reachability mask m_{Γ} restricts positive support to the vicinity

of Γ , preventing inaccessible surface regions from earning contact reward. The final object field is

$$O(\mathbf{x}) = \min\{\psi(\phi_{\mathcal{B}}(\mathbf{x})), m_{\Gamma}(\mathbf{x})\}. \quad (3)$$

Taking the minimum preserves the negative interior penalty even away from Γ . The corresponding swept field is the binary union indicator $\chi_{\mathcal{S}}(\mathbf{x}) = \max_i \mathbf{1}[\phi_{\mathcal{Q}}(\mathbf{T}_i^{-1}\mathbf{x}) \leq 0]$, where the maximum implements (1) and ensures that overlapping probe poses mark each voxel only once. Our geometric score is

$$F(\mathbf{X}) = \int_{\mathbb{R}^3} O(\mathbf{X}^{-1}\mathbf{x}) \chi_{\mathcal{S}}(\mathbf{x}) d\mathbf{x}. \quad (4)$$

Near-boundary overlap raises F , deep interpenetration lowers it, and distant swept occupancy contributes zero. This single correlation integral encodes boundary complementarity and non-penetration without extracting contact points or assigning correspondences.

Because the construction requires only an occupancy or signed-distance query, any rigid tool can replace $\mathcal{Q}$, with its size and anisotropy entering the measurement directly through (1).

C. Global docking by correlation

At voxel spacing h , let $O_R[k]$ and $S[k]$ sample the rotated object field and swept occupancy $\chi_{\mathcal{S}}$, respectively. Here $\mathcal{F}$ and $\mathcal{F}^{-1}$ denote the 3D discrete Fourier transform and its inverse, $\odot$ denotes element-wise multiplication, and the overbar denotes complex conjugation. The fast Fourier transform (FFT) then evaluates the cross-correlation over every translation $\Delta\mathbf{k} \in \mathbb{Z}^3$:

$$\begin{aligned} C_R[\Delta\mathbf{k}] &= \sum_{\mathbf{k}} O_R[\mathbf{k}] S[\mathbf{k} + \Delta\mathbf{k}] \\ &= \mathcal{F}^{-1} \left(\mathcal{F}(S) \odot \overline{\mathcal{F}(O_R)} \right) [\Delta\mathbf{k}]. \end{aligned} \quad (5)$$

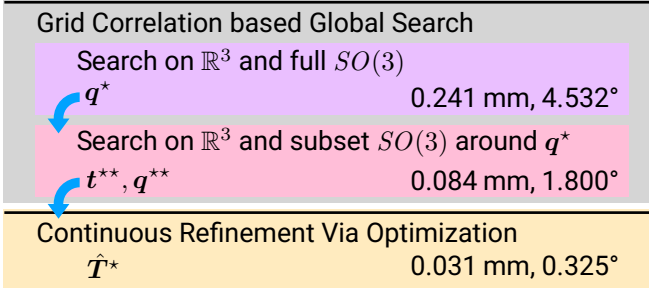


Fig. 3. Coarse-to-fine search: global $SE(3)$ docking, local $SO(3)$ resampling, and continuous $SE(3)$ refinement. Values are mean simulation errors.

With zero padding to prevent circular wrap-around, the FFT reduces translation search from quadratic in the number of voxels to $\mathcal{O}(M \log M)$ for M padded voxels.

We sample orientation with a low-discrepancy Super-Fibonacci sequence on $SO(3)$ [11]. For each orientation, Equation 5 returns its optimal translation, after which a second sequence restricted to the quaternion geodesic neighborhood of the best global sample improves angular resolution. The resulting pose X_0 initializes continuous pose refinement.

III. CONTINUOUS REFINEMENT VIA OPTIMIZATION

Because voxelization and sampled rotations bound the accuracy of global search, continuous refinement returns to the original geometry. For pose X , let $d_i(X) = \text{sdist}(XB, T_iQ)$ denote the minimum signed distance between the object and probe at trajectory sample i , with negative values denoting penetration. A smooth score $g(d)$ peaks at $d = 0$, vanishes beyond a positive clearance band, and decreases steeply under penetration. Using a bounded local pose parameterization $X(\xi)$ around X_0 , with $\xi \in \mathbb{R}^6$ and $\xi^\wedge \in \mathfrak{se}(3)$, we solve

$$\min_{\xi} J(\xi) = -\sum_{i=1}^N g(d_i(X(\xi))) + \lambda \|\xi\|_2^2, \quad (6)$$

subject to tight component-wise bounds on ξ . Rotational perturbations are applied about the object's geometric center, and every update remains on $SE(3)$.

Samples outside the clearance band contribute zero, while penetrating poses incur a strong penalty, so the objective does not force every trajectory sample onto the object surface.

The distance query returns an object-side witness point p_i and a consistent normal n_i . For an infinitesimal rigid perturbation $\delta\xi = [\delta t^\top, \delta\theta^\top]^\top$ at the current pose, the local signed-distance differential is

$$\begin{aligned} \delta d_i &\approx J_i \delta\xi, \\ J_i &= [n_i^\top \quad ((p_i - c) \times n_i)^\top], \end{aligned} \quad (7)$$

where c is the object center in the world frame. These contact sensitivities, together with $g'(d_i)$, are used to construct the refinement gradient. The complete finite-perturbation parameterization and its chain rule are given in the full paper [12].

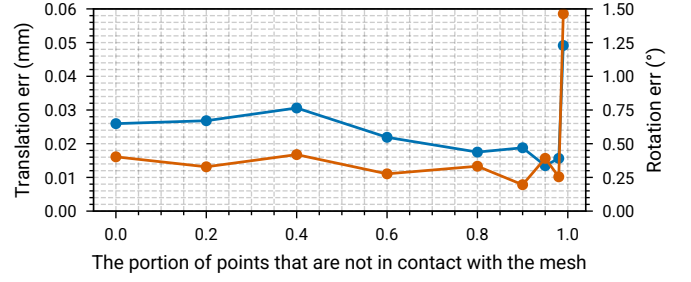


Fig. 4. Robustness to intermittent contact. Translation (blue) and rotation (orange) errors remain low with up to 98 % non-contact poses, but increase when no contact poses remain.

IV. EVALUATION

A. Simulation across diverse geometries

We generated three-direction contact sweeps on five meshes comprising a machined workpiece, MATLAB logo, toy car, front teeth, and molar teeth (Fig. 5), with independent 0.02 mm Gaussian position noise added to each trajectory.

Fig. 3 shows progressive error reduction across the three stages: correlation identifies the correct global basin, geodesic resampling reduces rotational quantization, and signed-distance refinement further improves the pose estimate. The final mean errors were 0.031 mm and 0.325°. To evaluate intermittent contact, we then retained increasing fractions of non-contact poses. Such poses would be outliers under a point-measurement model, yet in the swept representation they still impose non-penetration constraints without being assigned surface correspondences. Registration remained accurate when the non-contact fraction reached 98 % (Fig. 4), demonstrating the empirical benefit of the representation under intermittent contact.

B. Robot tooth preparation

We evaluated the method on a UR3e tooth-preparation robot, where a 1 mm-diameter cylindrical bur served as the probe (Fig. 1). A force–position controller maintained a 0.3 N contact force during an 80 s three-direction sweep, and registration used the calibrated bur geometry together with the recorded robot poses. The same robot then executed the planned preparation.

Because direct ground truth is difficult to obtain at this scale, we aligned pre- and postoperative intraoral scans using unchanged neighboring teeth and used the prepared tooth as an independent shape-consistency check. The estimated pose differed from this approximate ground truth by 0.42 mm and 3.75°, while designed-to-prepared surface errors ranged from −0.59 mm (over-cut) to 0.71 mm (under-cut). For comparison, we also evaluated an optical-tracker-based registration pipeline combining tracker measurements, hand–eye calibration, and a printed marker guide. In this trial, the resulting optical registration shifted the cutting trajectory and led to bur breakage after excessive contact, whereas preparation based on swept-volume registration preserved the neighboring teeth (Fig. 6).

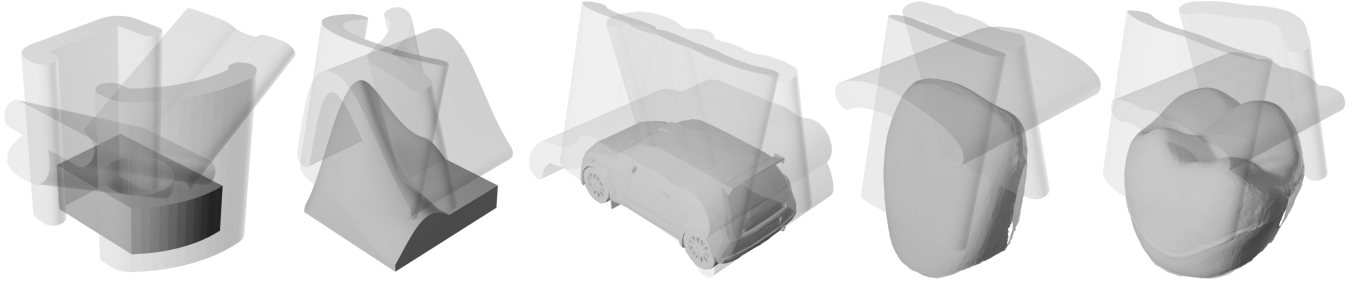


Fig. 5. Simulation overview. Contact trajectories were generated with a cylindrical probe on five object meshes. Translucent swept volumes show the simulated probe motion used for registration. Gaussian noise was added during the experiments but is omitted here for clarity.

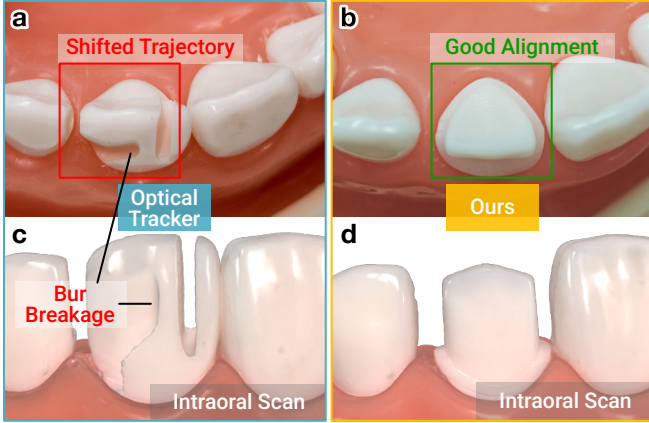


Fig. 6. Tooth-preparation comparison. The optical-tracker-based pipeline shifted the trajectory (left), whereas swept-volume registration produced the intended alignment (right).

V. DISCUSSION AND CONCLUSION

The contribution is a geometric representation of contact trajectories in which each recorded pose contributes the full probe body at its measured configuration. Their union preserves probe scale and orientation and encodes non-penetration constraints along the recorded motion. Contact poses support boundary agreement, while non-contact poses remain informative without being assigned surface correspondences. Together, saturation of repeated coverage in the union indicator and suppression of distant occupancy by the narrow-band score explain the low registration error observed with 98% non-contact poses. Fourier correlation and Lie-group refinement operate on the same representation during both global search and continuous refinement.

The current formulation assumes a known rigid object and probe, sufficiently accurate robot kinematics and tool-center calibration, and quasi-static contact. Robot pose errors deform the swept volume and therefore directly affect registration accuracy, while deformation, friction-dependent probe deflection, and object-model error remain outside the present model. The current implementation also runs offline and takes 115 s on average.

The most relevant extensions include uncertainty-aware geometric fields, symmetry-aware multi-hypothesis estimation, and active next-sweep selection, which together could provide

a common geometric representation for contact sensing, collision reasoning, and manipulation. Overall, treating the swept probe as the measurement turns contact registration into geometric inference on $SE(3)$, with the simulation and tooth-preparation experiments showing that this representation supports accurate registration under intermittent contact.

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